



Worm Brook 2025

A report on behalf of the Wye Salmon Association.

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A year in the life of the Worm Brook, as evidenced by citizen science data recorded during 2025.

Introduction

This report reviews citizen science water quality data obtained from the Worm Brook during 2025¹ by the Dore Citizen Science (CS) Team. It follows on from an earlier report (Green, 2024b) which examined the 2023-4 data. The earlier report provided a brief geographical overview of the sub catchment which I shall not repeat that here. A map is provided in Appendix A which shows the locations of the sites monitored. In general, the emphasis of this report will be to avoid presenting trend plots which can be viewed in [WyeViz](#) but rather attempt to provide a deeper analysis of what the data tells us.

In addition to the more common CS parameters (phosphate², TDS/SC³, pH & temperature), 2025 is the first full year of nephelometric turbidity data across the Worm sub catchment; ammoniacal nitrogen and nitrate/nitrite have also been introduced. These new parameters will be examined in this report.

2025 is notable for having had a long, dry summer. As we shall see, the river was on its bones from mid-year until well into the autumn and this has had a significant impact on water quality, particularly in the vicinity of point sources. Also of note was a severe pollution event in late March, discovered by a Dore Team member on a sampling visit to the middle Worm, in which it is believed that anaerobic digestate poured into the Knolton Brook for several hours. The EA attended site that day and subsequently reported finding dead fish in the Worm downstream of the Knolton Brook confluence.

Worm Brook sub catchment

Sampling locations

Sampling locations, as mapped in Appendix 1, are summarised in table 1 below. The Knolton Brook site (WSA 157) was added in June 2025, following the pollution event.

Reference	Site name	Description
<i>Allensmore Brook:</i>		
WSA 030	Worm, The Tram	Allensmore Brook, above Worm confluence
<i>Upper Worm (above the Allensmore confluence):</i>		
WSA 129	Worm, below Wormlow STW	Above Allensmore confluence, below Much Dewchurch STW
<i>Lower Worm (from the Allensmore confluence down to the Monnow confluence):</i>		
WSA 031	Worm, Gough's Bridge	Above Kilpeck STW
WSA 142	Worm, below Kilpeck STW	Below Kilpeck STW
WSA 143	Worm, Wormbridge	Below Wormbridge STW
WSA 032	Worm, Sawmills Footbridge	Most downstream location
<i>Jury Brook:</i>		
WSA 156	Worm, Jury Brook	Jury Brook, 1km above Worm confluence
<i>Knolton Brook:</i>		
WSA 157	Worm, Knolton Brook	Knolton Brook, 0.5km above Worm confluence (new in 2025)

Table 1: summary of WSA sampling sites in the Worm sub catchment.

Point sources

Three Dwr Cymru sewage treatment works are also shown on the map in Appendix 1; these are summarised in table 2 below. As noted in the earlier report, Much Dewchurch is of particular concern. It is at capacity and has high CSO (meaning screened only) emissions. During 2023, this CSO discharged for 6 weeks continuously. Moreover, it discharges into a small stream offering poor dilution. It is scheduled for some improvements under AMP8, including P-stripping. Nevertheless, members of the Dore Team have rightly been campaigning for a wetland to be constructed at this site. The other two STWs are

¹ CS data has been downloaded directly from the [EpiCollect](#) platform.

² In this report, all Hanna phosphate readings (mgPO₄/l) have been converted to RP (mgP/l).

³ TDS (total dissolved solids) and SC (specific conductivity) are essentially the same measurement, being related by a simple calibration factor. In this report I shall use SC, the electrical conductivity at 25°C.

somewhat smaller and are much further down the sub catchment, thus discharging into larger flow volumes. Nevertheless, as we shall see, they do have a measurable incremental impact in water quality.

Site name	Capacity (PE)	Input load (PE)
Much Dewchurch (new) STW	499	499
Kilpeck STW	n/a	80
Wormbridge STW	n/a	50

Table 2: summary of DC-WW sewage treatment plants (2024 Fol req.).

Flow monitoring

Unfortunately, the Worm Brook has no flow monitoring. In the 2023-4 report, we showed that the EA gauging station at Marstow Mill on the neighbouring Garren Brook was an acceptable substitute for data analysis purposes – the Worm and Garren sub catchments having reasonably similar topography and agriculture. I shall again make use of Marstow Mill flow data in this report, and a chart of the 2025 daily average discharge (volumetric flow rate) is included as Appendix B.

Results and discussion

Summary statistics

The data comprises 284 sets of readings across the 8 sites. Most sites have approximately weekly data throughout 2025, although Knolton Brook only commenced sampling in June. Sawmills footbridge, with only 4 samples covering the period January to March, will not be analysed. WSA 143 at Wormbridge is our most downstream site this year.

Annual statistics for the key parameters are shown in table 3.

Reference	Site name	Turbidity (NTU)	EC (uS/cm)	pH	RP (mgP/l)	TON (mgN/l)	Ammon. N (mgN/l)
WSA 030	Worm, The Tram	13	707	7.68	0.24	8.78	1.81
WSA 129	Worm, below Wormlow STW	13	615	7.71	0.59	6.92	0.17
WSA 031	Worm, Goughs Bridge	7	743	7.46	0.18	7.68	0.79
WSA 142	Worm, below Kilpeck STW	9	744	7.45	0.27	8.00	0.32
WSA 143	Worm, Wormbridge	15	636	7.71	0.25	6.92	0.26
WSA 156	Worm, Jury Brook	12	600	7.68	0.08	8.34	0.11
WSA 157	Worm, Knolton Brook	6	683	7.43	0.26	7.78	0.57

Table 3: annual mean values over 2025 for turbidity, electrical conductivity, pH, reactive phosphorus, total oxidised nitrogen; ammoniacal nitrogen is expressed as a 90%ile, in line with WFD requirements; high outliers are shown in red, low outliers in blue.

Compared to 2023-4 results, EC and pH are broadly similar. Numerically slightly different, probably reflecting the drier summer, but showing a similar geographical variation. There is nothing to note at this level of generalisation.

Turbidity as an annual mean is not terribly informative; I shall examine this in more detail below. RP, on the other hand, has increased massively, essentially reflecting the low flows over the summer. Water Framework Directive (WFD) 'good' status is site specific but, for the Worm generally falls around 0.07 mgP/l. Other than the Jury Brook which scores a 'moderate', all sites are a factor of 3 or so higher – well into WFD 'poor' status. The Worm below Wormlow STW (known by DC-WW as Much Dewchurch STW) is particularly bad.

The new parameter total oxidised nitrogen (TON), most of which is nitrate, is fairly uniform across the sub catchment. These numbers are not dissimilar to values on the neighbouring Gamber, being typical of areas with significant arable farming. Although there is no WFD requirement for TON, EA informal guidance (at the Wye Alliance/EA liaison meeting of 16-Jan-25) is that a value of circa 1.5 mgN/l would be appropriate. All sites are a factor of 4 to 5 above this.

The other new parameter, ammoniacal nitrogen presents a mixed picture with only four sites comfortably complying with the WFD 'good' status limit of $<0.6 \text{ mgN/l}$ (90%ile). Three sites, highlighted blue, actually get into WFD 'high' status ($<0.3 \text{ mgN/l}$). The high outlier is the Allensmore at the Tram which falls into 'poor' ($>1.1 \text{ mgN/l}$). Something bad seems to have occurred here.

As last year, the Jury Brook looks notably healthier than other areas of the Worm sub catchment based on these annual statistics. We shall now explore individual parameters in more detail. In doing so, I shall attempt to summarise by way of exemplars, rather than including charts of all parameters at all sites.

Turbidity

The pattern of turbidity data is closely reminiscent of results reported previously (Green, 2024a) for the Garren/Gamber sub catchment, showing distinctly different behaviour at low to moderate flows than at high flows. As the first exemplar, figure 1 shows the turbidity trend plot for WSA129 on the Worm below Much Dewchurch STW.

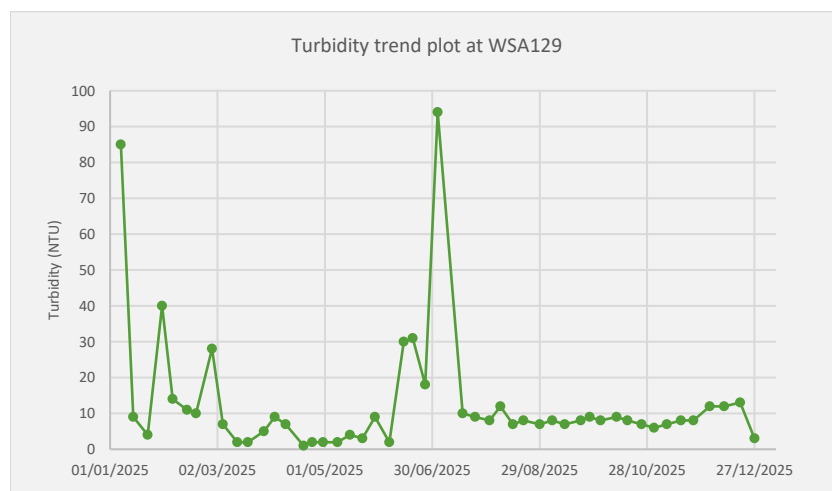


Figure 1: turbidity trend plot for WSA129, the upper Worm, about 2km below Much Dewchurch STW.

The three spikes at the beginning of the year correlate with the discharge spikes seen in Appendix B. But what occurred in the middle of the year (mid-June to mid-July), during dry weather? Examination of the descriptive fields in EpiCollect reveals this to be due to an ongoing algal bloom, generally described as 'greenish' or 'brownish'. The spike at 95 NTU is huge. This location remained murky for the remainder of the summer (compare to the levels in early May) but did not spike again. The patterns all become clearer when we correlate this parameter with discharge, as in figure 2 below.

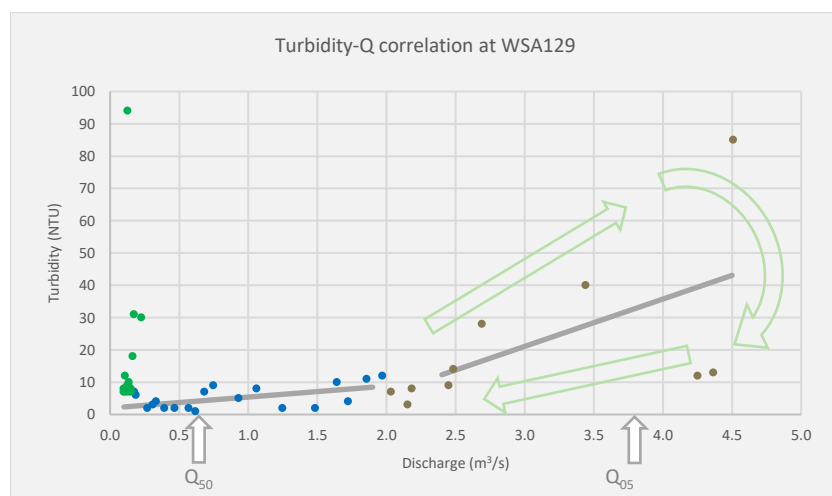


Figure 2: turbidity-discharge correlation for WSA129. Low-flow regime up to 2x annual average flow (blue), high-flow regime above 2x annual average flow (brown) and algal bloom (green).

Arrows beneath the axis indicate Q_{50} at $0.63 \text{ m}^3/\text{s}$ and Q_{05} at $3.7 \text{ m}^3/\text{s}$.

Indicative green arrow show the turbidity hysteresis: high turbidity on a rising river, lower on a falling river.

Below about $2 \text{ m}^3/\text{s}$, the data (blue points) are a good fit to a linear regression model (grey line) with a small positive gradient. Turbidity at flows below $0.2 \text{ m}^3/\text{s}$ should clearly be in low single figures. But the

high outliers, shown in green, all correspond to the algal bloom and the ongoing murkiness for the rest of the summer. Statistically, the green dots do not belong to the population of blue dots corresponding to the regression model.

Above 2 m³/s, we enter the high flow region where we find the hysteresis pattern noted in the Garren report (Green, 2024a). Turbidity is much higher on a rising river than on a falling river at the same flow – hence the large variance found in the data in this region. But we can again fit a linear regression which will allow us to characterise this region in terms of simple parameters. The value of the regression at Q₀₅ is a reasonable summary parameter for the high flow region. Likewise, the low flow region will be summarised by the regression value at Q₅₀. We shall also make use of the minimum observed turbidity.

Next, we shall look at a site downstream, WSA143 at Wormbridge. Figure 3 shows the turbidity – discharge correlation for this site.

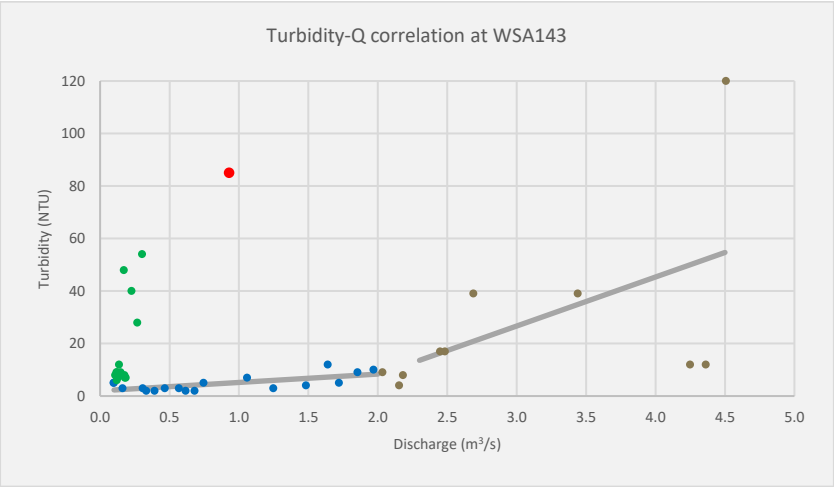


Figure 3: turbidity-discharge correlation for WSA143 at Wormbridge. Low-flow regime up to 2x annual average flow (blue), high-flow regime above 2x annual average flow (brown) and algal bloom (green).

The red datapoint corresponds to the Knolton Brook pollution incident on 28-March.

Once more we find a strong algal bloom lasting throughout June; two weeks earlier than the one upstream at WSA129. Peak turbidity at this time was similarly high, and once more the river remained murky (circa 8 NTU) after the bloom, compared to circa 3 NTU seen throughout May. The 85 NTU spike shown in red corresponds to the Knolton Brook pollution event on 28th March. Excluding the red and green points, the ‘normal’ low and high flow regimes are similar to the behaviours seen at WSA 129.

Rather than present a large number of charts⁴, we can summarise the turbidity data using the parameters described above.

Reference	Site name	Turbidity (NTU)			Algal bloom	
		Minimum (NTU)	Q ₅₀ (NTU)	Q ₀₅ (NTU)	Start date	Peak turb. (NTU)
WSA 030	Worm, The Tram	1	4	33	14-Jun	72
WSA 129	Worm, below Wormlow STW	1	4	31	14-Jun	94
WSA 031	Worm, Goughs Bridge	2	5		23-Aug	12
WSA 142	Worm, below Kilpeck STW	2	8		17-Jul	11
WSA 143	Worm, Wormbridge	2	4	40	29-May	54
WSA 156	Worm, Jury Brook	1	2	16	29-May	82
WSA 157	Worm, Knolton Brook	1	8		17-Jul	12

Table 4: key parameters summarising turbidity data; minimum is the lowest value recorded, Q₅₀ is the low flow regression value at 50%ile flow, Q₀₅ is the high flow regression value at 5%ile flow (blanks are sites with insufficient high flow data). All sites had algal blooms; start date and peak turbidity are shown here.

Looking at the ‘minimum’ column, we see that all sites are capable of running clean occasionally. But the Q₅₀ column shows us that only Jury Brook runs consistently clean at lower flows. Four sites show intermediate values (4-5), with two sites having values of 8. WSA142 may be impacted by its close

⁴ I have all charts for all sites and would be happy to supply these should anyone has a specific interest.
Page 5 of 15 ©2026 WSA

proximity to the Kilpeck sewage outflow. Overall, the Worm compares favourably with the Garren/Gamber where Q_{50} turbidity is generally over 10 NTU throughout the sub catchment⁵. Looking at Q_{05} , we see three sites all fairly similar and the Jury Brook standing out as much cleaner at high flows. 30-40 NTU is reasonably similar to the adjacent Gamber, and lower than the Garren. Missing values are where there are too few high flow measurements to analyse.

Algal blooms occurred at **all** monitored sites in 2025, with four of these being severe. An interesting observation is that these four were relatively early blooms, commencing in May/June and with an obvious correlation with solar input. For example, figure 3 shows the water temperature trend on the Jury Brook (other sites are similar). The blooms which commenced later in the summer (mid-July/August), were possibly triggered by other factors, but in any event were all less severe.

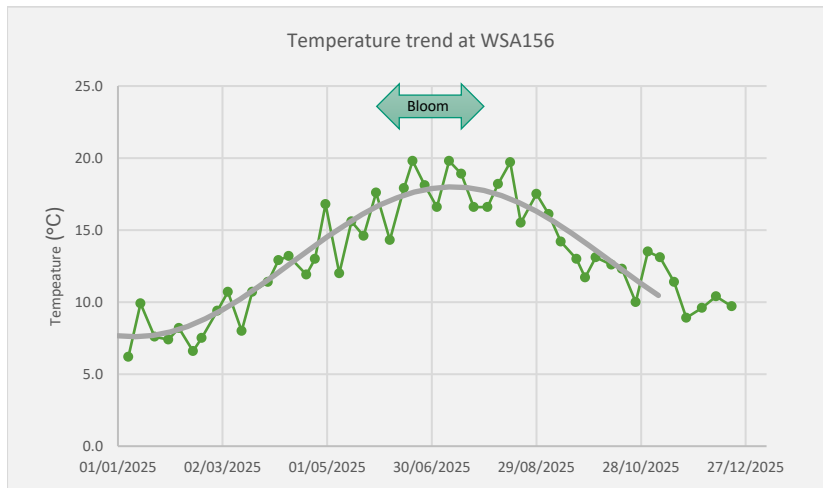


Figure 3: temperature trend on the Jury Brook, site WSA156. The arrow indicates the approximate duration of the algal bloom at this site.

The grey curve represents a best-fit sine function which is a reasonable approximation to the underlying trend.

Specific conductivity & pH

Both SC and pH are to a large extent driven by geology; hence annual trends are of limited utility as water quality indicators. The sedimentary bedrock in this area is high in CaCO_3 which dissolves readily in groundwater and hence dominates the conductivity of the river water. Most of what we measure as conductivity is due to dissolved mineral ions⁶, much of which is CaCO_3 . Nutrients such as nitrate and phosphate represent tiny additional contributions which are not discernible in normal conditions. The high CaCO_3 load (known as alkalinity) in the Worm also tends to increase pH. The 7.4-7.8 range seen in table 1 is typical of alkaline rivers.

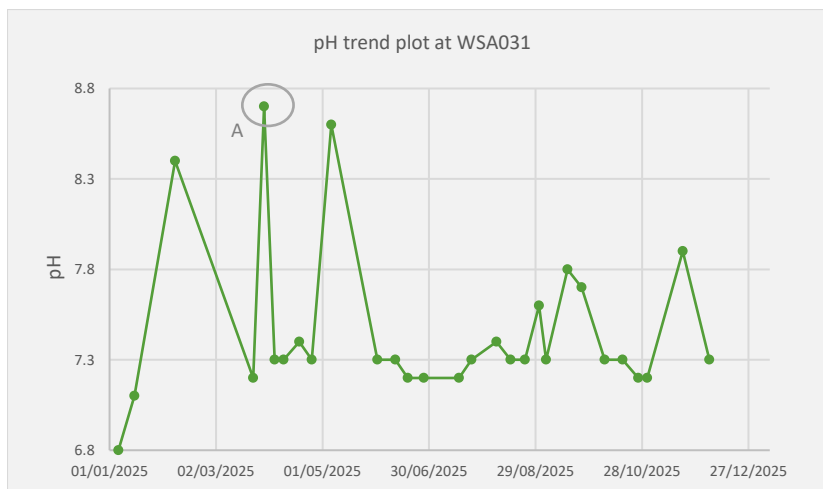


Figure 4: pH trend plot for Gough's Bridge, WSA031. There are six obvious positive transients and one negative one (06-Jan). The event circled at 'A' was on 29-Mar, the day after the Knolton Brook pollution.

⁵ The Garren has different geology which is thought to release more ultra-fine clay particles into the river.

⁶ In wet weather, when fast, surface pathways come into play, water has less overall contact time with soil and bedrock; this causes a gradual fall in conductivity at higher river flowrates.

Rather, the motivation for monitoring SC and pH lies in their ability to flag up acute pollution; we are essentially looking for transient 'spikes'. As an exemplar, figure 4 shows the pH trend plot for Gough's Bridge, WSA031. The quiescent state appears to be within the range 7.2 to 7.4. But there are also six notable positive (alkaline) transients and one negative (acidic) transient in early January. In an agricultural context, a potential cause of a negative transient is slurry which, unless chemically modified, is naturally highly acidic. However, this one on 06-Jan corresponds to a high flow event and is most probably explained by reduced CaCO_3 load. The most likely cause of alkaline pollutant is liquid digestate and, indeed, the event circled at 'A' on figure 4 was recorded the day after the Knolton Brook digestate pollution, Gough's Bridge being about 1.5km downstream of the Knolton Brook confluence. The cause of the similar spike on 6-May is unknown.

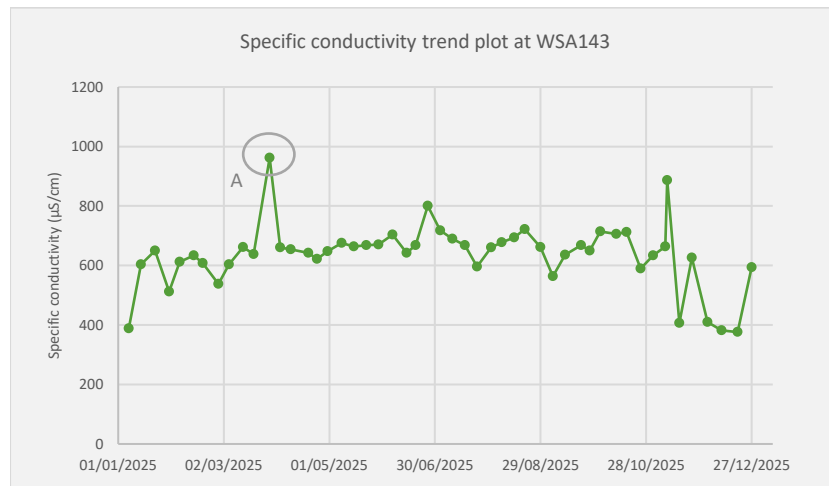


Figure 5: SC trend plot for Wormbridge, WSA143. There are Two obvious positive transients. The event circled at 'A' was on 28-Mar, the day of the Knolton Brook pollution.

Figure 5 shows the specific conductivity trend at Wormbridge, WSA143, which is located about 2km below Gough's Bridge. There are two notable positive transients. The first of these, labelled 'A' again corresponds to the Knolton Brook event on 28-Mar. The cause of the second one on 9-Nov is unknown. The negative excursions are generally associated with high flow, the one on 7-Jan confirming our diagnosis of the pH dip seen at Gough's Bridge the same day (figure 4).

The two exemplars above, being typical of data throughout the catchment, show that the value of these parameters lies not in their general trend, but rather in the transient outliers which are indicative of acute pollution events.

Ammoniacal nitrogen

Ammoniacal nitrogen is a composite of two chemical species: the ammonium ion NH_4^+ and unionised ammonia NH_3 , the latter being highly toxic to fish and other aquatic lifeforms. These are not independent pollutants but exist together in equilibrium, the ratio between them being determined by temperature and pH. Increasing pH and temperature both drive an increase in the fraction of the toxic unionised form. It is for this reason that the WFD limits for this substance are stated as 90%ile rather than annual average values. It is the peak values which have the potential to kill things, not the background average.

As an exemplar, figure 6 shows the ammoniacal nitrogen trend at Wormbridge, WSA143. General values are low, being mostly below the WFD 'good' status limit of 0.6 mgN/l. However, there are two notable transients. The one labelled 'A' is, once again, the Knolton Brook episode on 28-Mar. Note that the Hanna meter maxed out on this reading, so we have no idea how high the true reading was on this occasion. The other transient, on 14-Jan has no obvious explanation. It does not, for example, correspond to a high flow event.

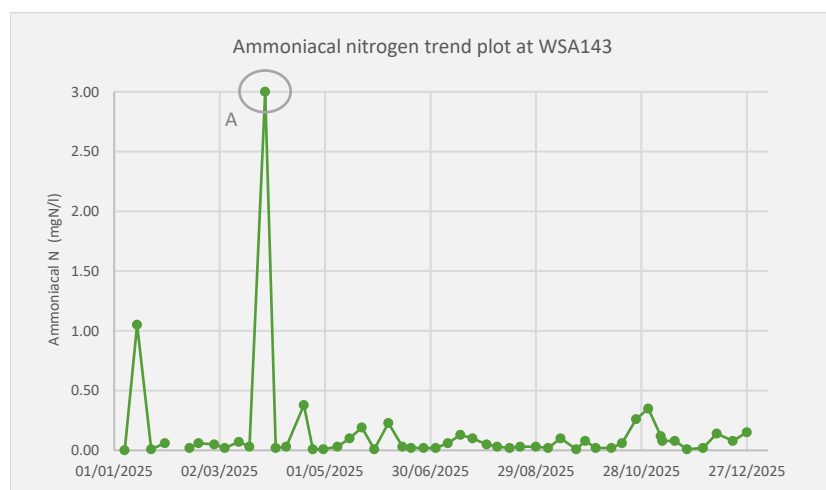


Figure 6: ammoniacal nitrogen trend plot for Wormbridge, WSA143. There are Two obvious positive transients. The event circled at 'A' was on 28-Mar, the day of the Knolton Brook pollution.

The above example is broadly typical of most of the Worm sites. Generally low readings punctuated by transient spikes, which based on the Knolton Brook episode, are indicative of acute pollution events. We can analyse this further, as summarised in table 5 below.

Reference	Site name	90%ile (mgN/l)	WFD status	No. of transients	Maximum (mgN/l)	Base 90%ile (mgN/l)
WSA 030	Worm, The Tram	1.81	Poor	19	3.00	0.17
WSA 129	Worm, below Wormlow STW	0.17	High	0	0.32	0.17
WSA 031	Worm, Goughs Bridge	0.79	Moderate	2	1.09	0.13
WSA 142	Worm, below Kilpeck STW	0.32	Good	1	0.87	0.28
WSA 143	Worm, Wormbridge	0.26	High	2	3.00	0.19
WSA 156	Worm, Jury Brook	0.11	High	1	0.79	0.10
WSA 157	Worm, Knolton Brook	0.57	Good	1	0.61	0.46

Table 5: summary statistics for ammoniacal nitrogen; high outliers are shown in red, low outliers in blue.

The first column, '90%ile', reproduces the data from table 3. The next column shows the appropriate WFD classification for these values. On this basis, five of our seven sites would meet the national objective of good or better status. The next two columns show the number of transients exceeding 0.6 mgN/l (this being the target for 'good' status), followed by the highest peak value observed, some of which are alarmingly high numbers⁷. In the final column we re-calculate the 90%ile for the datasets excluding the identified transients. The point of doing this is to show that six sites (in blue) would conform to WFD 'high' status, but for a handful of 'accidental releases' of something nasty. Background levels of ammoniacal nitrogen across the Worm catchment are not that bad. The problem to solve is transient spikes.

But something very odd happened on the Allensmore Brook at The Tram over the summer. Figure 7 shows the trend plot at this location. For the first six months of the year, we see consistently low values compliant with WFD 'high' status. At the end of June someone started doing something. And carried on doing it until the first autumn rains arrived in mid-November. This emission effectively continued throughout the period of lowest flow on the river, the driest period of the summer. It is clearly anthropogenic and is not fixed flow, in the sense that it doesn't follow a dilution curve. It's either on, albeit with different values each day, or its off. This all looks like irrigation run-off, and, given its proximity, Allensmore Nurseries must be a prime suspect. Checking abstraction licenses on the [EA Water Abstraction Licenses](#) website shows two ground water abstraction point close by, both operated by Allensmore Nurseries under license numbers WA/055/0017/012 and 19/55/17/0438. A couple of miles upstream at Green Farm near Allensmore, Owens & Sons also have a groundwater abstraction

⁷ Note that the 3.00 mgN/l values represent maxed-out Hanna meters; the true values are potentially much higher.

point under license number 19/55/17/0454. Clearly someone is spraying so much water onto crops that it's draining into the river taking fertiliser with it. In which case, we ought to see a similar pattern on other nutrients (see RP below).

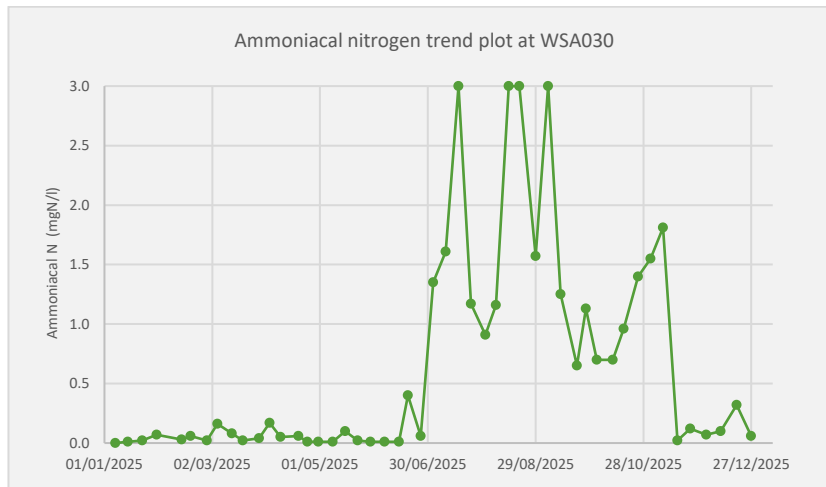


Figure 7: ammoniacal nitrogen trend plot for The Tram, WSA030. The apparent fall in mid-September is due to slightly increased flowrate at this time, rather than a fall in output.

Reactive phosphorus

In the summary statistics presented in table 3, we saw that WSA129, below Much Dewchurch STW had very high RP levels, so let's start there. Figure 8 shows the trend plot for this site.

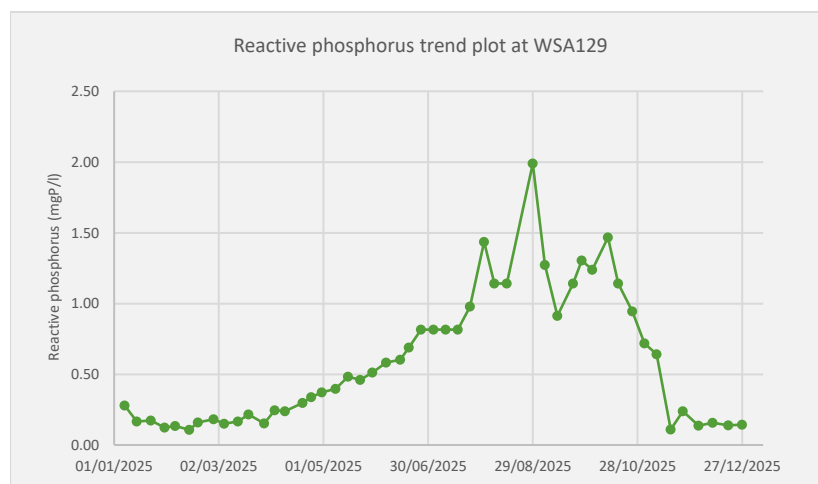


Figure 8: RP trend plot for WSA029, below Much Dewchurch STW.

During the early part of the year, RP values hovered around 0.12 mgP/l. From the beginning of April, levels rose progressively, suggesting a classic dilution curve. The four flat points in early July represent the maximum reading of a Low range Hanna checker (0.82 mgP/l = 2.5ppm PO₄). Subsequently, a high range checker was obtained, allowing the remainder of the summer to be properly captured.

In the 2023-4 data we found a very clear fixed flow behaviour at this site corresponding to 1.3 kgP/day of RP discharging from Much Dewchurch STW located a little over 2km upstream. Figure 8 shows the RP-discharge correlation for the 2025 data. We then use the solver tool in Excel to find the best fit to a classic load apportionment model using the following equation:

$$RP = \frac{A}{Q} + B \cdot Q^{D-1}$$

where Q is discharge, A is mass flow of RP from the STW, B & D are parameters of the diffuse agricultural RP contribution. A, B & D are all optimised by the solver to best fit the data. This tells us that the STW emitted about 1.2 kgP/day of RP; very similar to the 1.3 kgP/day we estimated from the 23-24 dataset. At the median flowrate, Q₅₀ (0.61 m³/s in figure 9), 85% of RP at this site comes from the STW.

At slightly below Q_{95} flowrate, where the rivers ended up in late August, this proportion rose to around **99%**. Note that the extreme low flows experienced in 2025 were less than half of the corresponding 2024 minimum flows, which will clearly double the impact of fixed flow sources such as STWs.

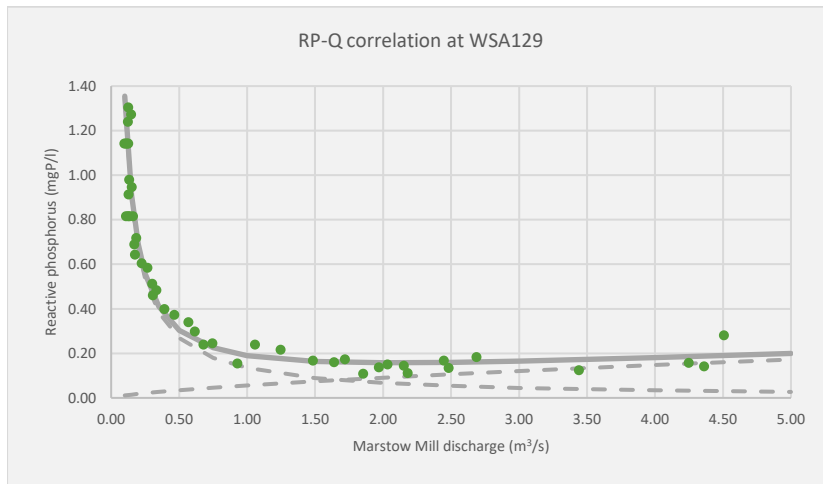


Figure 9: RP-discharge correlation for WSA029, below Much Dewchurch STW.

Solid grey line is the best-fit LAM curve. Dashed lines represent the two terms of the LAM model, the falling $1/Q$ dilution curve of the STW contribution and the rising diffuse agriculture signal.

Discharge on the Worm at WSA129 is estimated at $0.1 \times$ the Marston Mill discharge.

At the other extreme, we find Jury Brook with almost no fixed flow component. Figure 10 shows the RP-discharge plot for this site. The tiny low-flow upswing corresponds to a mass flow of 0.04 kgP/day – equivalent to the septic tank load from circa 25 people – entirely plausible.

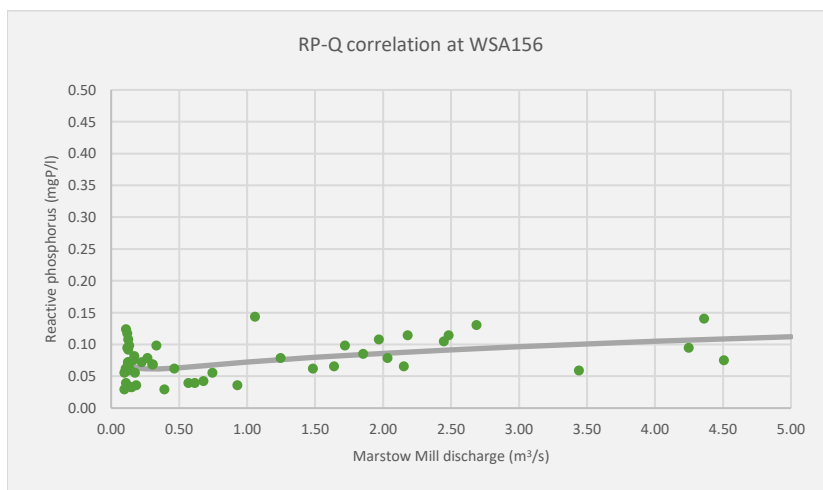


Figure 10: RP-discharge correlation for Jury Brook, WSA156. Note the change of scale from figure 9.

Solid grey line is the best-fit LAM curve.

Discharge on the Worm at WSA129 is estimated at $0.1 \times$ the Marston Mill discharge.

A summary of the LAM analysis of RP data for the Worm sites is shown in table 6 below. For reference, the first column shows the annual mean RP values reproduced from table 3. 'Fixed flow' is the first term of the LAM model. 'FF fraction at Q_{50} ' expresses this as a fraction of the best-fit total RP load at median flow. We have not presented fixed flow parameters for The Tram; something odd occurred at this location over the summer which we shall discuss below.

Reference	Site name	Mean RP (mgP/l)	Fixed flow (kgP/day)	FF fraction at Q_{50}	Diffuse at Q_{50} (mgP/l)
WSA 030	Worm, The Tram	0.24	Fixed flow not valid		0.05
WSA 129	Worm, below Wormlow STW	0.59	1.21	85%	0.06
WSA 031	Worm, Goughs Bridge	0.18	1.34	52%	0.05
WSA 142	Worm, below Kilpeck STW	0.27	1.74	51%	0.06
WSA 143	Worm, Wormbridge	0.25	1.99	54%	0.06
WSA 156	Worm, Jury Brook	0.08	0.04	11%	0.06
WSA 157	Worm, Knolton Brook	0.26	Not analysed due to insufficient data		

Table 6: summary statistics for RP and the RP LAM analysis. Annual mean RP is also reproduced from table 1.

The first observation, examining the 'Diffuse at Q₅₀' column, is the notable consistency across the sub catchment. Agricultural diffuse pollution seems to provide a very consistent background RP level at all locations. Unfortunately, this alone is sufficient to take annual mean RP up to or a little above the limit for FWD 'good' status. Comparing the 'Mean RP' and 'FF fraction at Q₅₀' columns, we see that the considerable variability in mean RP across the sub catchment is being driven primarily by the high fixed-flow contributions. The sole exception, Jury Brook, having a very small fixed-flow load is the only location coming anywhere close to the WFD target. We have already discussed the dire situation at WSA129, caused by Much Dewchurch STW, but the whole middle Worm downstream of this point is blighted by fixed-flow emissions. In the 'Fixed flow' column, we see how the P load from Much Dewchurch is further topped up by 0.4kgP/day from Kilpeck STW and a further 0.2kgP/day from Wormbridge STW.

Returning now to The Tram, WSA030, figure 11 shows the RP trend plot for this location. As expected, this strongly mimics the ammoniacal nitrogen trend shown in figure 7.

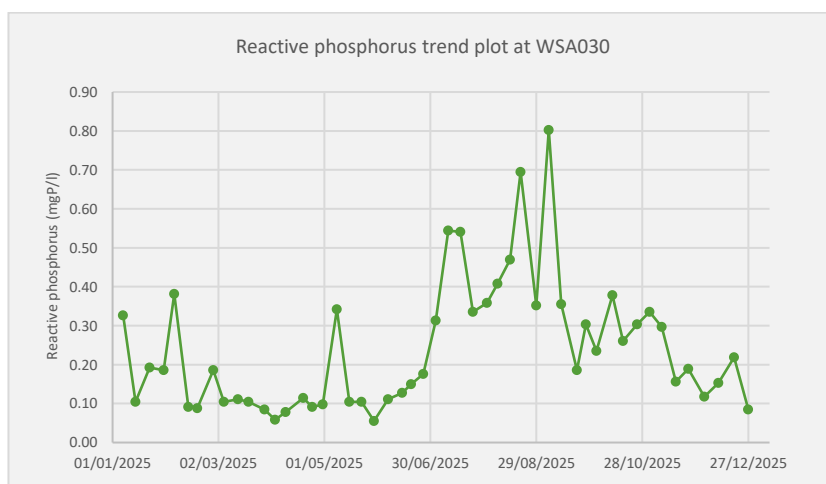


Figure 11: RP trend plot for The Tram, WSA030.

Rather than a progressive ramp-up through the summer, in response to falling flowrate (as seen in figure 8), we find a step change at the beginning of July. This feature is categorically not fixed flow dilution. Just like figure 7, it persisted until 11-November. It also exhibits the partial drop around 12-September (caused by a slight flow increase at that point). This strongly supports the earlier suggestion regarding an irrigation related causation. It should also be noted that a similar pattern was not found in our analysis of 2023-4 data. Nor was there any suggestion of a significant fixed flow RP component at this location.

Total oxidised nitrogen

TON is measured using test strips and, because of the coarse resolution inherent in this method, options for analysis are limited to basic trending. By way of a cross reference, EA monitoring in the Worm sub catchment in 2025 has occurred at three sites⁸: 3 samples on the lower Worm above the Dore confluence, 4 samples from the upper Worm at Lowe Bridge Farm (a little upstream of WSA129), and 3 samples from the Jury Br. at Wormbridge Mill (a little below WSA156). Each of these sites showed a similar pattern, beginning at 3-5 mgN/l in January and rising to around 7 mgN/l in the growing season.

Figure 12 shows the trend plot for WSA129. Overlaid in orange are the temporally sparse but high accuracy data reported by EA. Within the resolution limits of the test strips, there is a reasonably good correlation with the EA data. Although not presented here, the Jury Brook data is also in good quantitative agreement with EA determinations. Beyond these comparisons, there is nothing to add regarding TON, beyond the earlier discussion of the annual average values.

⁸ Data downloaded from the [EA water quality explorer](#).

The test strips also report nitrite, which represents a tiny fraction of TON. But I have not attempted any specific analysis of nitrite here.

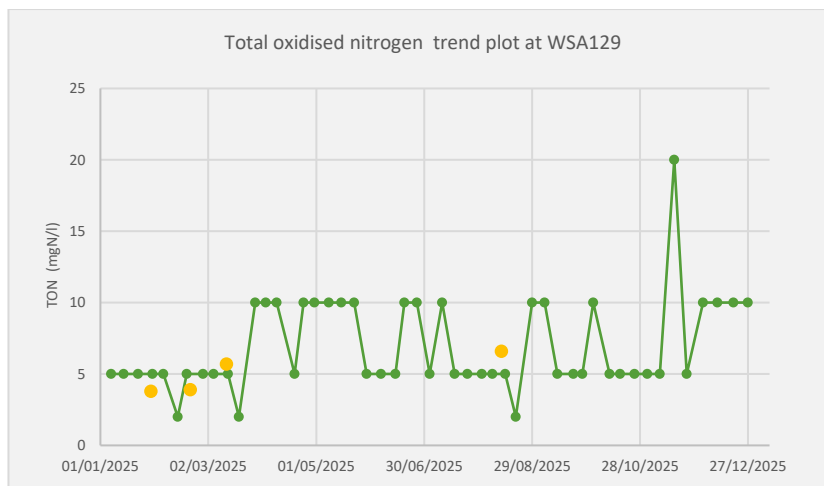


Figure 12 TON trend plot for WSA029, below Much Dewchurch STW. Orange circles are EA laboratory determinations from sampling at Lowe Bridge Farm, a few hundred metres upstream.

The Knolton Brook pollution

Given the severity of the Knolton Brook event, it is useful to review what was detected by CS data at the time.

The first relevant data was taken at 11:00am from WSA 142 below Kilpeck STW, and approximately 1.8km downstream of the Knolton confluence. This indicated a significant turbidity spike of 66 NTU, consistent with the reported visual appearance and smell. There was also a large spike on SC, rising from circa 640 up to 942 $\mu\text{S}/\text{cm}$ but, surprisingly, no significant change was observed on pH. On the nutrient parameters, there was a big spike in RP, with the Hanna phosphate checker maxing out at 0.82 mgP/l; ammoniacal nitrogen and TON were not measured at that time, priority being given to reporting the incident to EA. Given the volumetric flowrate on that day, a back of the envelope calculation suggests that transit time down to Kilpeck was probably around 2.5 hours, so this dreadful leak had clearly been discharging for quite some time before its discovery. At 15:11 that day, the river was sampled further downstream at Wormbridge, WSA 143, showing a similar pattern. Spikes on turbidity (85 NTU), SC (962 $\mu\text{S}/\text{cm}$), maxed out meters on both RP and ammoniacal nitrogen (3 mgN/l), but nothing unusual on TON or pH. During that afternoon, EA attended site, identified the source and stopped the leak. The next morning at 11:43, some limited measurements were taken at Goughs Bridge, WSA 031, about 1.5 km downstream of Knolton Brook. SC had partially settled back down to 750 $\mu\text{S}/\text{cm}$ and turbidity was visually less pronounced but not measured. RP also had fallen back to 0.11 mgP/l. But the curious thing is that pH was now significantly elevated at 8.7 – exactly what you would expect of a digestate leak.

To summarise, turbidity, conductivity, phosphate and ammoniacal N, all showed large upswings during the pollution. Notably, pH only indicated a problem the day after.

Conclusion

- RP
 - The dry summer of 2025 saw the Worm fall to flowrates significantly below Q_{95} during August; this being less than half the lowest flowrates seen in 2024. The extremely high RP values seen in 2025 were caused by reduced dilution of fixed-flow sources (STWs) in these lower flows. LAM analysis suggests that RP mass flows from the three STWs was essentially unchanged from 2024.
 - LAM analysis shows that the diffuse agricultural component of RP is fairly uniform throughout the catchment. It is sufficient by itself to push annual mean RP above the WFD

target, as witnessed by the marginal failure this year of Jury Brook which has minimal fixed-flow load. The fact that all other sites have annual mean RP circa 3x the WFD target is a consequence of the STWs within the sub catchment.

- Turbidity
 - All sites have been shown to be capable of achieving low turbidity (0-2 NTU) on occasion. In the low flow regime, all sites have reasonably low average turbidity at Q_{50} flowrate; noticeably lower than in the adjacent Gamber for example. In the high-flow regime, most sites analysed had average turbidity at Q_{05} flowrate similar to the Gamber. The Jury Brook stands out as having the lowest average turbidity, both at Q_{50} and at Q_{05} , perhaps a consequence of its more energetic flow.
 - During 2025 all sites had discernible algal blooms which badly degraded low flow turbidity for part of the year. At four sites the turbidity increase was severe with values rising into the 50 – 100 NTU range around June (i.e. peak sunlight, not peak nutrients). These were typically followed by ongoing lower levels of turbidity (circa 8 NTU) for the rest of the summer.
- Ammoniacal nitrogen
 - This has proved a useful parameter. General background levels of ammoniacal N are comfortably below WFD targets. However, transient spikes have occurred at most sites which push the specified 90%ile metric above the WFD limit. These transients are presumably associated with localised sloppy agricultural practices.
- Total oxidised nitrogen
 - TON readings appear uniform across the sub catchment, with growing-season levels in the 5-10 mgN/l region, typical of other areas with significant arable agriculture (e.g., Gamber).
 - Test strip data are consistent with recent EA laboratory determinations of this parameter.
- SC & pH
 - Annual mean values are similar to 2023-4. The main utility of these parameters lies in their ability to pick up transient pollution events, of which there were many. Again, presumably the result of sloppy agricultural practices.
- Knolton Brook
 - The acute pollution incident on the Knolton Brook was discovered and notified to EA by our CS volunteers. Proving the importance of 'eyes on the river'.
 - This event caused noticeable increases in RP, ammoniacal N, SC and turbidity at all downstream sites. Oddly, pH did not respond at the time but was elevated the next day.
- Allensmore Brook
 - A serious level of nutrient pollution (RP & ammoniacal N) was observed in this stream between 3-July, and 8-Nov. RP was suddenly elevated from circa 0.1mgP/l in June to 0.3-0.8 mgP/l throughout this period. For ammoniacal N, the increase was more dramatic, switching from circa 0.07 mgN/l throughout the spring, to a minimum of 0.65 mgN/l, with multiple peaks exceeding the range of the Hanna meter at 3.0 mgN/l.
 - 3.0 mgN/l would imply unionised ammonia peaking in the region of 0.08mgN/l (at pH7.8 at 22°C) which is almost certainly in the acute toxicity range for fish.
 - We note a groundwater abstraction point (license 19/55/17/0438) for the purposes of direct spray irrigation, operated by Allensmore Nurseries, and situated adjacent to their glass houses, immediately upstream of our monitoring site. It seems eminently plausible that drainage from this facility is going directly into the brook.
 - The Environment Agency need to inspect this site as a matter of urgency.

Appendix A: monitoring site locations & point sources

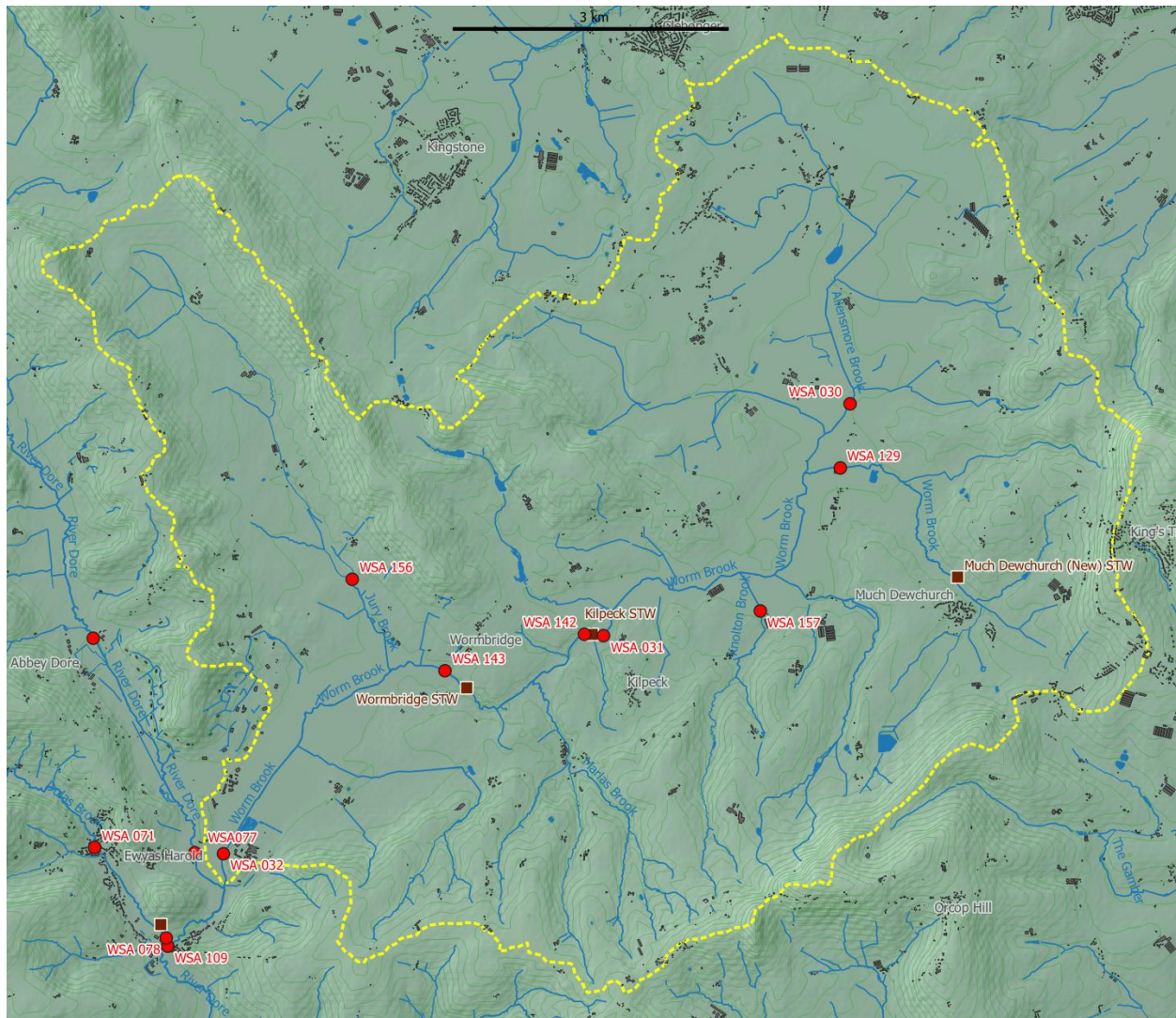


Figure A1: the Worm sub catchment showing monitoring sites (red circles) and STWs (brown squares); WSA 157 commenced in June 2025, and WSA 032 has very limited data. Note the steeper-sided Jury Brook valley, which has more similarity to the adjacent Dore than to the gentler topography of the Worm.

Appendix B: Marstow Mill discharge

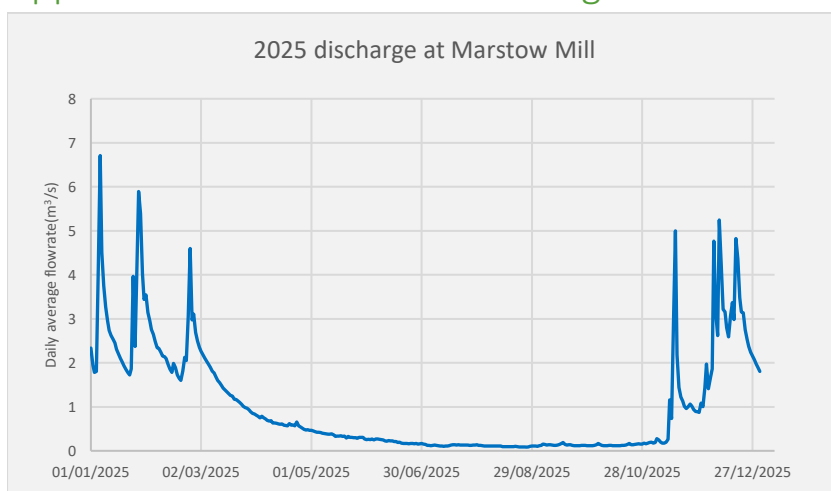


Figure B1: daily average discharge (volumetric flowrate) at Marstow Mill on the Garren Brook. Being in an adjacent and very similar sub catchment, this monitoring station has been found to be a good substitute for discharges in the Worm sub catchment after linear adjustment for catchment areas and average rainfall. For 2 weeks spanning 13-28th August, the discharge was consistently below Q_{95} – which did not occur in 2023 or 2024. (Data downloaded from [DEFRA Hydrology Data Explorer](#)).

References

Green, G. (2024a). *Garren turbidity 2023-4*.

Green, G. (2024b). *Worm Brook 2023-24*.

Abbreviations

CS	Citizen science
EA	Environment Agency
EC	Electrical conductivity: what the HM Digital EC & TDS meters measure; this quantity being temperature dependent (also see SC).
LAM	Load apportionment model: an analytical model of nutrient pollution, separating the effects of fixed flow sources and diffuse agricultural sources.
RP	Reactive phosphorus: the phosphorus fraction reported by the Hanna meter and by laboratory determinations; this approximates to orthophosphate only, other 'condensed' forms not being detected.
SC	Specific conductivity: the electrical conductivity referenced to a standard temperature of 25°C (also see TDS).
STW	Sewage treatment works
TDS	Total dissolved solids: a measure of the amount of dissolved material in the water, derived from SC using a material-specific calibration factor. The factor used on the HM digital TDS meter is 0.5 for NaCl. So 2x the TDS reading gives SC.
TON	Total oxidised nitrogen, essentially nitrate plus nitrite
WFD	Water Framework Directive; the EU legislation underpinning UK regulations for surface water quality.

Acknowledgements

Huge appreciation is due to the Dore citizen scientists who have diligently collected this data, going out in all weathers on a weekly basis, for no other reason than they care deeply about the sad plight of our rivers. Their dedication is admirable. I hope that this short report does justice to their efforts and helps clarify the issues which their data exposes.

Thanks are also due to Michael Carpenter for his ceaseless work maintaining EpiCollect and WyeViz and to the rest of the team on the Wye Alliance Data Steering Group.

In addition to our CS data, this report makes use of Ordnance Survey mapping data, plus various DEFRA/EA datasets available online.